

Day: 19

* The Solo Investigation Mindset

- Working on practical things, this is what a real SOC analyst does on their work phase.

* The Complete Investigation Checklist

- Use this every single time you analyse a PCAP:
 - Protocol Hierarchy → unexpected protocols?
 - Conversations → most active IPs?
 - IO Graph → beaconing pattern?
 - DNS Filter → DGA? NXDOMAIN? Suspicious TLDs?
 - HTTP Filter → POST? Unusual UA? /gate.php?
 - Export HTTP Objects → malware dropped?
 - Follow TCP Stream → read actual conversation
 - TLS Filter → SNI reveals C2 domain?
 - Network Miner Hosts → OS fingerprinting
 - Network Miner Files → Files extracted
 - Network Miner Creds → stolen credentials?

- Zeek dns.log → NXDOMAIN count
- Zeek conn.log → longest connections
- Zeek http.log → POST requests

* Indicators of Compromise (IOC) Table

- When you find something suspicious, record it like this:

Type	Value	Tool Found	Confidence
IP	185.220.X.X	Wireshark Conversations	High
Domain	xkgZz3mpa.ru	DNS Filter	High
Port	4444	TCP Filter	Medium
URL	/gate.php	HTTP Filter	High
Hash	d41d8c...	Network Miner Files	High
UA	Mozilla/4.0	HTTP Filter	Medium

* What Makes a Good Investigation

- Speed matters: Real analysts triage fast — Protocol Hierarchy and Conversations first, always. These give

you the big picture in 60 seconds.

- Evidence chain: Every finding needs a filter, packet number, or log line backing it up. "I think this IP is suspicious" is not evidence. "Conversations tab shows 185.220.x.x sent 2.3MB in 47 connections over 4 minutes" Is evidence.

- Content matters: Not every ~~post~~ POST is malicious. Not every unknown domain is DGA. Ask:

1) Is this normal software phoning home?

2) Is this a CDN or cloud service?

3) Does the timing pattern make sense for normal traffic?

- The write-up comparison: After reading the write-up, the most important question is not "did I get it right?" but "why did I miss what I missed?" That gap is exactly what to focus on next.

* Common Mistakes Analysts Make

1) Jumping to Filters too fast

→ Always do Protocol Hierarchy first

2) Ignoring DNS

→ DNS reveals everything — always check it

3) Not checking User - Agent

→ Malware always has unusual UAs

4) Trusting HTTPS = safe

→ Check SNI — domain is still visible

5) Missing beaconing

→ Use IO Graph — it shows timing visually

6) Not using NetworkMiner

→ It finds credentials you'll miss in WireShark

LAB — The Full Solo Investigation:

1. Go to <https://malware-traffic-analysis.net>
2. Download a PCAP exercise

2024-08-15-traffic-analysis-exercise.pcap

File Edit View Go Capture Analyze Statistics Telephony Wireless Tools Help

Apply a display filter ... <Ctrl-/>

No.	Time	Source	Destination	Protocol	Length	Info
1	0.000000	0.0.0.0	255.255.255.255	DHCP	383	DHCP Request - Transaction ID 0xf96e0aac
2	0.000755	10.8.15.4	255.255.255.255	DHCP	342	DHCP NAK - Transaction ID 0xf96e0aac
3	0.010733	0.0.0.0	255.255.255.255	DHCP	342	DHCP Discover - Transaction ID 0x29b669b8
4	0.011654	10.8.15.4	255.255.255.255	DHCP	353	DHCP Offer - Transaction ID 0x29b669b8
5	0.023612	0.0.0.0	255.255.255.255	DHCP	389	DHCP Request - Transaction ID 0x29b669b8
6	0.024230	10.8.15.4	255.255.255.255	DHCP	358	DHCP ACK - Transaction ID 0x29b669b8
7	0.039631	Intel_03:54:82	Broadcast	ARP	42	Who has 10.8.15.1? Tell 10.8.15.133
8	0.039716	Cisco_02:a7:4b	Intel_03:54:82	ARP	42	10.8.15.1 is at 00:16:9c:02:a7:4b
9	0.048765	10.8.15.133	224.0.0.22	IGMPv3	54	Membership Report / Join group 224.0.0.251 for any sources
10	0.057770	Intel_03:54:82	Broadcast	ARP	42	Who has 10.8.15.133? (ARP Probe)
11	0.057853	10.8.15.133	224.0.0.22	IGMPv3	54	Membership Report / Join group 224.0.0.251 for any sources
12	0.072370	10.8.15.133	224.0.0.22	IGMPv3	54	Membership Report / Join group 224.0.0.252 for any sources
13	0.077461	10.8.15.133	224.0.0.22	IGMPv3	54	Membership Report / Leave group 224.0.0.252
14	0.077704	10.8.15.133	224.0.0.22	IGMPv3	54	Membership Report / Join group 224.0.0.252 for any sources
15	0.079539	10.8.15.133	224.0.0.251	MDNS	81	Standard query 0x0000 ANY DESKTOP-H8ALZBV.local, "QM" question
16	0.079799	10.8.15.133	224.0.0.251	MDNS	91	Standard query response 0x0000 A 10.8.15.133
17	0.081786	10.8.15.133	224.0.0.252	LUNMR	75	Standard query 0x67ad ANY DESKTOP-H8ALZBV
18	0.358379	Intel_03:54:82	Broadcast	ARP	42	Who has 10.8.15.4? Tell 10.8.15.133
19	0.358694	Dell_8c:9c:40	Intel_03:54:82	ARP	42	10.8.15.4 is at 64:00:6a:8c:9c:40
20	0.358829	10.8.15.133	10.8.15.4	DNS	99	Standard query 0xa098 SRV _ldap._tcp.dc._msdcs.lafontainebleu.org
21	0.359155	10.8.15.4	10.8.15.133	DNS	169	Standard query response 0xa098 SRV _ldap._tcp.dc._msdcs.lafontainebleu.org SRV 0
22	0.381043	10.8.15.133	10.8.15.4	DNS	94	Standard query 0xc32 A win-jegjix7q9rs.lafontainebleu.org
23	0.381332	10.8.15.4	10.8.15.133	DNS	110	Standard query response 0xc32 A win-jegjix7q9rs.lafontainebleu.org A 10.8.15.4
24	0.390455	10.8.15.133	10.8.15.4	LDAP	272	searchRequest(1) "<ROOT>" baseObject

> Frame 1: Packet, 383 bytes on wire (3064 bits), 383 bytes captured (3064 bits)
> Ethernet II, Src: Intel_03:54:82 (00:1c:bf:03:54:82), Dst: Broadcast (ff:ff:ff:ff:ff:ff)
> Internet Protocol Version 4, Src: 0.0.0.0, Dst: 255.255.255.255
> User Datagram Protocol, Src Port: 68, Dst Port: 67

0000 ff ff ff ff ff ff
0010 01 71 17 37 00 00
0020 ff ff 00 44 00 43
0030 0a ac 00 00 00 00

1. Write down every protocol you see and its percentage. Any unexpected protocols?

Wireshark - Protocol Hierarchy Statistics - 2024-08-15-traffic-analysis-exercise.pcap

Protocol	Percent Packets	Packets	Percent Bytes	Bytes	Bits/s	End Packets	End Bytes	End Bits/s	PDU/s
Frame	100.0	18189	100.0	11674125	47 k	0	0	0	18189
Ethernet	100.0	18189	2.2	254646	1038	0	0	0	18189
Address Resolution Protocol	0.4	81	0.0	2268	9	81	2268	9	81
Internet Protocol Version 4	99.6	18108	3.1	362192	1476	0	0	0	18108
Internet Control Message Protocol	0.1	11	0.0	1646	6	11	1646	6	11
Internet Group Management Protocol	0.0	8	0.0	128	0	8	128	0	8
Transmission Control Protocol	84.5	15367	2.7	317856	1295	11650	243516	992	15367
Data	0.9	163	23.7	2767966	11 k	163	2767966	11 k	163
Distributed Computing Environment / Remote Procedure Call (DCE/RPC)	1.9	344	0.8	98041	399	102	45515	185	344
Active Directory Replication	0.8	138	0.1	17072	69	138	17072	69	138
DCE/RPC Endpoint Mapper	0.3	52	0.1	10184	41	52	10184	41	52
Local Security Authority	0.1	12	0.0	1680	6	12	1680	6	12
Microsoft Network Logon	0.1	10	0.0	2788	11	10	2788	11	10
SAMR (pidl)	0.2	30	0.0	2250	9	30	2250	9	30
Hypertext Transfer Protocol	3.6	658	5.9	683326	2785	339	450472	1836	658
Line-based text data	1.7	309	1.6	187960	766	309	187960	766	309
Media Type	0.0	8	1.6	186244	759	8	186244	759	8
Online Certificate Status Protocol	0.0	1	0.0	1782	7	1	1782	7	1
Kerberos	0.3	50	0.6	66948	272	50	66948	272	50
Lightweight Directory Access Protocol	1.4	256	1.5	171972	701	256	129687	528	271
NetBIOS Session Service	1.5	270	0.6	73043	297	0	0	0	270
SMB (Server Message Block Protocol)	0.0	3	0.0	207	0	3	207	0	3
SMB2 (Server Message Block Protocol version 2)	1.5	267	0.6	71756	292	231	56932	232	289
Data	0.0	4	0.0	3946	16	4	3946	16	4
Transport Layer Security	11.0	2009	42.6	4967420	20 k	2009	3579443	14 k	2206
User Datagram Protocol	15.0	2722	0.2	21776	88	0	0	0	2722
Connectionless Lightweight Directory Access Protocol	0.2	36	0.1	7965	32	36	7965	32	36
Domain Name System	2.3	419	0.3	38785	158	419	38785	158	419
Dynamic Host Configuration Protocol	0.0	6	0.0	1915	7	6	1915	7	6
Link-local Multicast Name Resolution	0.0	1	0.0	33	0	1	33	0	1
Multicast Domain Name System	0.0	5	0.0	208	0	5	208	0	5
NetBIOS Datagram Service	0.0	4	0.0	328	1	0	0	0	4
SMB (Server Message Block Protocol)	0.0	4	0.0	476	1	0	0	0	4
SMB MailSlot Protocol	0.0	4	0.0	100	0	0	0	0	4
Microsoft Windows Browser Protocol	0.0	4	0.0	132	0	4	132	0	4
NetBIOS Name Service	0.1	12	0.0	816	3	12	816	3	12
Network Time Protocol	0.0	6	0.0	720	2	6	720	2	6
QUIC IETF	11.8	2151	15.7	1835864	7484	2151	1834466	7479	2154
Simple Service Discovery Protocol	0.5	82	0.1	13818	56	82	13818	56	82

- QUIC, TLS, HTTP, DCE/RPC, LDAP, SMB2, Kerberos, DNS

2. Sort by bytes – which IP has the most traffic? Write down the top 3 IPs.

- TCP 84.5% → UDP 15.0% → QUIC 11.8% → TLS 11.0% → HTTP 3.6% → DNS 2.3%.

3. Look for:

- Random-looking domain names (DGA)
- Repeated queries to same domain (beaconing)
- NXDOMAIN responses (DGA failures)
- Unusual TLDs (.ru, .xyz, .tk)

No.	Time	Source	Destination	Protocol	Length	Info
472	2.897512	10.8.15.133	10.8.15.4	DNS	94	Standard query 0x7262 A win-1001X7Q9RS.lafontainebleu.org
438	2.656591	10.8.15.4	10.8.15.133	DNS	190	Standard query response 0xc3a SRV _ldap._tcp.Default-First-Site-Name._sites.lafontainebleu.org SRV 0 100 389 win-jeglix7qrs.lafontainebleu.org A 10.8.15.4
437	2.656163	10.8.15.133	10.8.15.4	DNS	120	Standard query 0xc3a SRV _ldap._tcp.Default-First-Site-Name._sites.lafontainebleu.org
372	2.539594	10.8.15.4	10.8.15.133	DNS	205	Standard query response 0x4635 SRV _ldap._tcp.Default-First-Site-Name._sites.DomainDnsZones.lafontainebleu.org SRV 0 100 389 win-jeglix7qrs.lafontainebleu.org A 10.8.15.4
371	2.539398	10.8.15.133	10.8.15.4	DNS	135	Standard query 0x4635 SRV _ldap._tcp.Default-First-Site-Name._sites.DomainDnsZones.lafontainebleu.org
319	2.385482	10.8.15.4	10.8.15.133	DNS	205	Standard query response 0xa48e SRV _ldap._tcp.Default-First-Site-Name._sites.ForestDnsZones.lafontainebleu.org SRV 0 100 389 win-jeglix7qrs.lafontainebleu.org A 10.8.15.4
318	2.385334	10.8.15.133	10.8.15.4	DNS	135	Standard query 0xa48e SRV _ldap._tcp.Default-First-Site-Name._sites.ForestDnsZones.lafontainebleu.org
125	2.172140	10.8.15.4	10.8.15.133	DNS	204	Standard query response 0xe824 SRV _kerberos._tcp.Default-First-Site-Name._sites.dc._msdcs.lafontainebleu.org SRV 0 100 88 win-jeglix7qrs.lafontainebleu.org A 10.8.15.4
124	2.171895	10.8.15.133	10.8.15.4	DNS	134	Standard query 0xe824 SRV _kerberos._tcp.Default-First-Site-Name._sites.dc._msdcs.lafontainebleu.org
115	2.145915	10.8.15.4	10.8.15.133	DNS	204	Standard query response 0x2c47 SRV _kerberos._tcp.Default-First-Site-Name._sites.dc._msdcs.lafontainebleu.org SRV 0 100 88 win-jeglix7qrs.lafontainebleu.org A 10.8.15.4
114	2.145622	10.8.15.133	10.8.15.4	DNS	134	Standard query 0x2c47 SRV _kerberos._tcp.Default-First-Site-Name._sites.dc._msdcs.lafontainebleu.org
111	2.115215	10.8.15.4	10.8.15.133	DNS	200	Standard query response 0xbef5 SRV _ldap._tcp.Default-First-Site-Name._sites.dc._msdcs.lafontainebleu.org SRV 0 100 389 win-jeglix7qrs.lafontainebleu.org A 10.8.15.4
110	2.114931	10.8.15.133	10.8.15.4	DNS	130	Standard query 0xbef5 SRV _ldap._tcp.Default-First-Site-Name._sites.dc._msdcs.lafontainebleu.org
73	1.540676	10.8.15.4	10.8.15.133	DNS	171	Standard query response 0xdca2 A go.microsoft.com CNWE go.microsoft.com.edgeskey.net CNWE e11290.dspg.akamaiedge.net A 23.33.138.184
72	1.500263	10.8.15.133	10.8.15.4	DNS	76	Standard query 0xdca2 A go.microsoft.com
62	1.260310	10.8.15.4	10.8.15.133	DNS	227	Standard query response 0xf65f A www.msftconnecttest.com CNWE ncsl-geo.trafficmanager.net CNWE www.msftncsl.com.edgesuite.net CNWE s1961.g2.akamai.net A 23.205.118.48 A 23.205.118.62
61	1.217879	10.8.15.133	10.8.15.4	DNS	83	Standard query 0xf65f A www.msftconnecttest.com
60	1.192079	10.8.15.4	10.8.15.133	DNS	164	Standard query response 0x238f No such name A upad.lafontainebleu.org SOA win-jeglix7qrs.lafontainebleu.org
59	1.191880	10.8.15.133	10.8.15.4	DNS	83	Standard query 0x238f A upad.lafontainebleu.org
56	1.174242	10.8.15.4	10.8.15.133	DNS	164	Standard query response 0xd296 No such name A upad.lafontainebleu.org SOA win-jeglix7qrs.lafontainebleu.org
55	1.173944	10.8.15.133	10.8.15.4	DNS	83	Standard query 0xd296 A upad.lafontainebleu.org
33	0.510398	10.8.15.4	10.8.15.133	DNS	169	Standard query response 0xbfe0 SRV _ldap._tcp.dc._msdcs.lafontainebleu.org SRV 0 100 389 win-jeglix7qrs.lafontainebleu.org A 10.8.15.4
32	0.509640	10.8.15.133	10.8.15.4	DNS	99	Standard query 0xbfe0 SRV _ldap._tcp.dc._msdcs.lafontainebleu.org
23	0.381332	10.8.15.4	10.8.15.133	DNS	138	Standard query response 0xc752 A win-jeglix7qrs.lafontainebleu.org A 10.8.15.4